

EC3218 — Portfolio Analysis

Summer 2025 Examination · Paper 1 · Full Marking Scheme

Module	EC3218 — Portfolio Analysis
Internal Examiner	Dr Hugh Farrell
Institution	University College Cork
Session	Summer 2025, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt 2 of 3 (100% of the paper). All questions carry equal weight. This written exam is worth 60% of the module; the remaining 40% is the S&P 500 assignment.
Materials	Non-programmable calculator permitted

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions.

Source basis. This scheme follows Dr Farrell's own EC3218 lecture material closely throughout — the January Effect (Rozeff and Kinney, 1976), the behavioural bias content, Roll's Critique, the Fama-French findings, the efficient frontier/CAL framework, frontrunning, rational bubbles, and the four performance measures are all drawn directly from the module's own slides and their cited sources, referenced at each question.

Section A · Question 1

The Efficient Market Hypothesis: seasonal anomalies, testing the forms, and behavioural biases — 100 marks

A1(a)

[30 marks]

The January Effect contends that stock returns are higher in January than in other months. If this effect is true, does it violate the Efficient Market Hypothesis? If so, which form does it violate?

A1(b)

[40 marks]

Suggest one way each to test the semi-strong form and the strong form of the Efficient Market Hypothesis.

A1(c)

[30 marks]

Explain how a conservatism bias, an overconfidence bias and regret avoidance may lead to deviations from the Efficient Market Hypothesis.

FULL MODEL ANSWER

Does the January Effect violate the EMH? Yes. If stock returns are genuinely, reliably higher in January than in other months, this is a predictable, repeating pattern in **past price/return data alone** — an investor could, in principle, earn abnormal returns simply by knowing what calendar month it is and adjusting their holdings accordingly, without needing any new information at all.

Which form does it violate? the weak form. The weak form of the EMH states that asset prices already incorporate all past price and volume information, meaning technical analysis — trading strategies based purely on historical price patterns — cannot generate abnormal returns. A reliable seasonal pattern like the January Effect (Rozeff and Kinney, 1976) is exactly this kind of pattern: it requires no private information and no analysis of public announcements, only knowledge of the calendar and historical returns data, so its existence is a direct violation of weak-form efficiency specifically (and, by extension, of the semi-strong and strong forms too, since weak-form efficiency is a necessary condition nested within both).

EXAM TECHNIQUE

Other seasonal anomalies documented in the same literature — the April Effect (Jegadeesh and Titman, 1993), the August Effect (Draper and Paudyal, 1997), and the Day-of-the-Week Effect (French, 1980) — are creditable alternative examples if a different seasonal effect is asked about, since they are all weak-form violations for exactly the same reason.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly concludes this violates the EMH AND correctly identifies the

weak form specifically **AND** correctly explains why (a purely past-price-based pattern requires no private or new public information to exploit).

15-22	15-22	Correctly identifies the weak form, but the reasoning for why is generic or incomplete.
8-14	8-14	Correctly concludes this violates the EMH but identifies the wrong form (semi-strong or strong).
0-7	0-7	Incorrectly concludes this does not violate the EMH, or no credible answer given.

Testing semi-strong form efficiency — the event study: an **event study** examines how quickly and completely stock prices respond to the public release of new information. The classic example from the module's own material is Busse and Green (2002), who examined stock returns in the minutes following analysts' stock discussions on CNBC — tracking cumulative returns second-by-second after the on-air mention. **What semi-strong efficiency predicts:** if the market is semi-strong form efficient, the full price adjustment should happen **essentially instantaneously** upon the information becoming public, with no further drift (up or down) afterward, since any residual, predictable drift would itself represent an exploitable, publicly-available pattern. Busse and Green found prices responded within **seconds** of the initial mention, with positive reports fully incorporated within about a minute — evidence broadly **consistent** with semi-strong efficiency.

Testing strong form efficiency — testing whether insiders earn abnormal returns: since strong-form efficiency requires that **even private information** is already reflected in prices, the natural test is to examine whether corporate **insiders** — who by definition have access to private information — consistently earn abnormal returns on their own trades, or whether outsiders who mimic insider trades (via mandatory disclosure filings) can profit from doing so. The module's own material cites several such studies: Jaffe (1974) found insiders possess genuinely special information, and that outsiders could profit by following disclosed insider trades; Pope et al. (1990) similarly found abnormal returns available from a trading strategy based on news of insider dealings; and Ziobrowski et al. (2011) found that members of the US House of Representatives earned abnormal returns on their own stock trades. **What this implies:** since these studies consistently find abnormal returns available to insiders (or those following them), this is evidence **against** strong-form efficiency.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly proposes an event study for semi-strong form AND correctly explains what result would confirm/reject efficiency AND correctly proposes an insider-trading-based test for strong form AND correctly explains what result would confirm/reject efficiency for each, ideally with a specific study cited for either.
21-30	21-30	Both correct test types proposed (event study; insider trading test), but the interpretation of what result would support/reject efficiency is not fully explained for one or both.
11-20	11-20	Only one of the two test types correctly proposed and explained.
0-10	0-10	No credible test proposed for either form.

Conservatism bias: investors update their beliefs **too slowly** in response to new evidence, meaning they initially **underreact** to news. Since semi-strong efficiency specifically requires that new public information is incorporated into

price **immediately**, a conservatism bias directly breaks this — the price only gradually drifts toward its new, correct level over time rather than jumping there at once, leaving a temporary, exploitable gap between price and fundamental value.

Overconfidence bias: investors **overestimate the precision of their own forecasts and abilities** — a well-documented pattern (as the module's material notes, 90% of Swedish drivers rank themselves as better-than-average). In investing, this manifests as excessive trading activity: Barber and Odean (2000) found that investors with higher trading activity perform significantly **worse** than passive investors, net of costs. This deviates from the EMH not by creating a pricing anomaly directly, but by showing that investor behaviour itself is systematically irrational in a way that generates real, avoidable costs — overconfident traders churn their portfolios chasing perceived edges that, on average, don't exist, which is difficult to reconcile with the EMH's assumption of rational, profit-maximising investors.

Price Response to Good News: Efficient vs. Under- vs. Overreaction

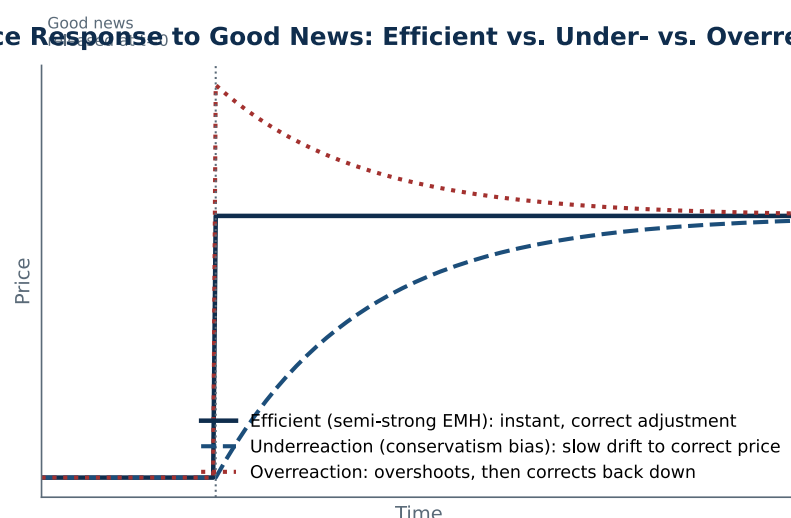


Fig. 1 — Semi-strong EMH predicts an instant, correct price adjustment to good news. Conservatism bias produces underreaction (a slow drift to the correct price); other biases can produce overreaction (an overshoot followed by a correction).

Regret avoidance: investors who make **unconventional** decisions that turn out badly blame themselves more than if a conventional decision turns out badly — losing money on an unknown start-up feels worse than losing the same amount on a blue-chip portfolio, even though the financial loss is identical. This can generate systematic mispricing: because it "takes more courage" (i.e. carries more regret risk) to invest in unconventional stocks — such as smaller firms or those with a high book-to-market ratio — investors may demand a **premium** to hold them, above and beyond what their actual risk justifies, which is one proposed behavioural explanation for the small-firm and value-stock anomalies (see Q. A2(b)) that is separate from Fama and French's own risk-based explanation for the same patterns.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	All three biases correctly explained AND each is explicitly connected to a specific way it breaks a specific form/prediction of the EMH (conservatism → semi-strong's instant-adjustment requirement; overconfidence → the rational-investor assumption via costly overtrading; regret avoidance → a behavioural premium on unconventional stocks) AND at least one specific study or example cited.
15-22	15-22	All three biases correctly explained, but the specific EMH-violation mechanism for at least one is only asserted rather than explained.

8-14 **8-14** Only two of the three biases correctly explained.

0-7 **0-7** Fewer than two biases correctly explained.

Question A1 Total: 100 Marks

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Q1

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Section A · Question 2

Roll's Critique, the Fama-French model, and the efficient frontier — 100 marks

A2(a)

[30 marks]

Briefly explain Roll's Critique of CAPM tests.

A2(b)

[40 marks]

Outline the Fama and French 3-factor model and explain how it differs from CAPM. Explain the findings of the model. Do these findings indicate that the Efficient Market Hypothesis is not correct? Explain your answer.

A2(c)

[30 marks]

What is meant by the term "efficient frontier"? Draw the Minimum Variance Frontier for a set of risky assets and mark the efficient frontier and the minimum variance portfolio. Use your diagram to explain why no rational investor would ever hold a portfolio not on the efficient frontier. Finally, draw the Capital Allocation Line and show the tangency portfolio. Explain the significance of this portfolio.

FULL MODEL ANSWER

The core problem: CAPM is stated in terms of **the** market portfolio, M — a portfolio containing every investable risky asset in existence, weighted by market value. In practice, **no such portfolio can actually be observed or traded** — researchers testing CAPM must substitute a proxy, such as the FTSE-100 or S&P 500, which captures only a subset of investable assets (typically large, publicly-listed domestic equities, excluding bonds, real estate, private equity, human capital, and international assets).

Why this makes CAPM untestable — the dual/joint hypothesis problem: any test of CAPM is **simultaneously** a test of (i) whether CAPM itself is true, and (ii) whether the chosen proxy is a reasonable stand-in for the true market portfolio. These two cannot be disentangled: the true market

portfolio M could be mean-variance efficient (CAPM true) while the proxy used in the test is not efficient, causing the test to **wrongly reject** a true model; or, conversely, the proxy could *appear* efficient (causing the test to seemingly confirm CAPM) even though the true, unobservable market portfolio is not — meaning CAPM would appear to hold in the test while actually being false. Because the test can never separate "CAPM is wrong" from "the proxy is wrong," Roll (1977) concluded that **CAPM in its pure form is not testable at all**.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly identifies that the true market portfolio is unobservable and proxies must be used AND correctly explains the dual/joint hypothesis problem (cannot separate "CAPM is false" from "the proxy is a bad stand-in") AND correctly concludes CAPM is therefore untestable in its pure form.
15-22	15-22	Correctly identifies the market-portfolio-proxy problem, but the specific joint-hypothesis reasoning for why this makes CAPM untestable (rather than just "harder to test") is not fully explained.
8-14	8-14	Vague reference to "we can't observe the true market" without the joint-hypothesis mechanism.
0-7	0-7	No credible explanation given.

Outlining the model: the Fama-French 3-factor model explains returns using **three** factors rather than CAPM's single beta: (i) the market factor (CAPM's own beta, $r_m - r_f$), (ii) **Small Minus Big (SMB)** — the excess return of small-cap stocks over large-cap stocks, and (iii) **High Minus Low (HML)** — the excess return of stocks with a high book-to-market ratio over those with a low book-to-market ratio (i.e. "value" stocks over "growth" stocks). Empirically, SMB is calculated as the return difference between a portfolio of the smallest 50% of firms by market cap and the largest 50%; HML is calculated as the return difference between the top 30% and bottom 30% of firms by book-to-market ratio.

How it differs from CAPM: CAPM asserts that **beta is the only variable** that should predict an asset's expected return — no other characteristic should matter once beta is accounted for. The Fama-French model directly tests this by adding SMB and HML as extra explanatory variables: if CAPM were true, both SMB and HML should carry **zero** risk premium in the data, since beta alone should already capture everything relevant.

The findings: using US stock data from 1963-1991, Fama and French found the 3-factor model **convincingly explained variation in returns** that CAPM alone could not — both SMB and HML carried statistically and economically significant positive premiums (small stocks outperformed large stocks; high book-to-market "value" stocks outperformed low book-to-market "growth" stocks), directly contradicting CAPM's prediction that these premiums should be zero. Fama and French (1998) later confirmed the value effect internationally, finding value stocks outperformed in 12 of 13 markets tested, with an average 7.68% return difference between low- and high-book-to-market portfolios.

Do these findings indicate the EMH is not correct? Not necessarily — this is genuinely contested, and the module's material presents both sides.

Position	Argument
Fama and French's own view (EMH defended)	SMB and HML are proxies for genuine, additional sources of risk that beta alone doesn't capture — small and value firms are riskier (e.g. more exposed to financial distress), so their higher average returns are simply fair compensation for that extra risk, not mispricing. On this reading, the findings are evidence <i>against CAPM specifically</i> , but not against the EMH more broadly, since prices can still be fully, rationally reflecting all available information even though a single-factor (beta-only) model fails to capture all the relevant risk.
Lakonishok, Shleifer and Vishny's view (EMH challenged)	LSV (1994) tested value vs. glamour stocks from 1963-1990 and found value stocks outperformed by 10-11% per year, while also finding no evidence value stocks did worse in "bad" economic states — the pattern a pure risk-based story would predict. If value/small stocks aren't actually riskier, their higher returns represent a genuine, exploitable market inefficiency (e.g. driven by regret avoidance or forecasting-error biases, see Q. A1(c)), which would be a direct violation of at least semi-strong form efficiency.

The resolution — this is exactly Roll's Critique again: notice that this debate has the same structure as Q. A2(a): to know whether SMB/HML premiums reflect risk or mispricing, you would need to know the *true* risk model — but since SMB/HML premiums are themselves estimated empirically rather than derived from first-principles theory, the same **joint hypothesis problem** applies: any test is simultaneously testing "is the market efficient" and "is this particular factor model the correct measure of risk," and the two cannot be cleanly separated.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly outlines the three factors and how SMB/HML are constructed AND correctly explains the CAPM contrast (beta-only vs. multi-factor) AND correctly states the empirical findings (significant SMB/HML premiums) AND gives a balanced EMH discussion covering both the risk-based (Fama-French) and mispricing-based (LSV) interpretations, ideally connecting back to the joint hypothesis problem.
21-30	21-30	Correctly outlines the model and findings, with a one-sided EMH discussion (only the risk-based or only the mispricing-based view given).
11-20	11-20	Model outlined correctly, but findings or EMH implications not meaningfully addressed.
0-10	0-10	No credible outline of the model given.

What the efficient frontier is: the **Minimum Variance Frontier (MVF)** shows, for every possible level of expected portfolio return, the **minimum** standard deviation (risk) achievable using some combination of the available risky assets. The **efficient frontier** is the **upper portion** of the MVF — specifically, the part running from the **minimum variance portfolio** (the single point on the MVF with the lowest possible risk of all) upward.

Minimum Variance Frontier, Efficient Frontier, and the CAL

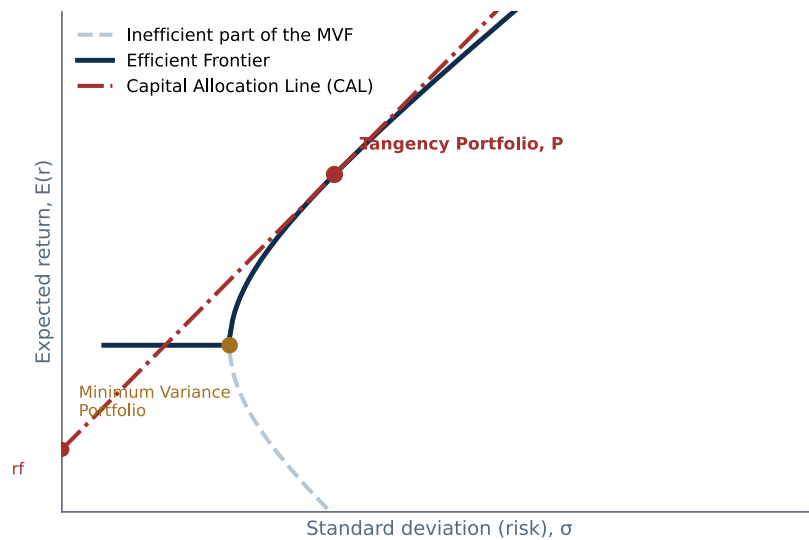


Fig. 1 — The Minimum Variance Frontier (both the solid efficient upper arc and the dashed inefficient lower arc), the minimum variance portfolio, and the Capital Allocation Line drawn tangent to the frontier at the tangency portfolio, P.

Why no rational investor would ever hold a portfolio below the minimum variance portfolio: for every point on the **lower**, inefficient arc of the MVF, there exists a portfolio directly above it on the efficient (upper) arc, achieved with the *exact same level of risk* (the same standard deviation) but offering *strictly higher expected return*. Since any rational investor prefers more expected return to less for a given level of risk, every portfolio on the lower arc is **dominated** — a rational investor would always switch to the efficient-frontier portfolio directly above it, which is strictly better on the one dimension (return) that differs, with nothing given up on the other (risk). This is why the efficient frontier, not the full MVF, is the relevant opportunity set for any rational, return-seeking investor.

The Capital Allocation Line and the tangency portfolio: once a risk-free asset is introduced, the investor's opportunity set expands further: the **Capital Allocation Line (CAL)** is drawn from the risk-free rate on the vertical axis, tangent to the efficient frontier at exactly one point — the **tangency portfolio, P**. The CAL *dominates every point on the efficient frontier itself* except at the single point of tangency, since combining the risk-free asset with P achieves a strictly better risk-return trade-off (a higher Sharpe ratio — the slope of the CAL) than any portfolio of risky assets alone could.

The significance of the tangency portfolio: P is the **unique** combination of risky assets that **every** rational investor should hold, **regardless of their individual risk tolerance** — this is the **separation property**: the choice of the optimal risky portfolio (a purely technical, objective problem — find the portfolio maximising the Sharpe ratio) is entirely separate from the investor's personal choice of how much to allocate between P and the risk-free asset (a subjective decision depending on individual risk aversion). A more risk-averse investor simply holds more of the risk-free asset and less of P; a less risk-averse investor holds more of P (potentially borrowing at the risk-free rate to hold more than 100% of P) — but every investor's risky holdings are the same portfolio, P, just scaled differently.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly defines the efficient frontier as the upper MVF arc from the minimum variance portfolio AND the diagram correctly shows the MVF, efficient frontier, and minimum variance portfolio AND the dominance

argument for why the lower arc is never rational is correctly explained
AND the CAL and tangency portfolio are correctly drawn and the separation property correctly explained.

15-22	15-22	Diagram and definitions correct, but either the dominance argument or the separation property explanation is missing or incomplete.
8-14	8-14	Diagram attempted with some correct elements (e.g. the MVF shape) but missing the efficient frontier distinction, minimum variance portfolio, or CAL/tangency portfolio.
0-7	0-7	No credible diagram or explanation given.

Question A2 Total: 100 Marks

Section A · Question 3

Frontrunning, rational bubbles, and portfolio performance measures — 100 marks

A3(a)

[30 marks]

Explain what is meant by "frontrunning" and why it is considered unethical.

A3(b)

[30 marks]

Explain what a financial bubble is. If you know that an asset is currently in a bubble, is it ever rational to hold that asset? Explain your answer.

A3(c)

[40 marks]

Discuss four measures of portfolio performance evaluation.

FULL MODEL ANSWER

What frontrunning is: frontrunning is a specific type of insider trading in which someone enters into a position based on advance knowledge that a **large transaction** is about to happen — one big enough to move the price — and profits from that anticipated price move. The classic case is a **broker** who receives a large client order (e.g. to buy a large block of shares): the broker buys the same shares for their own account first, then executes the client's order — which pushes the price up — and finally sells their own shares into the resulting price rise for a quick profit. A related form involves an analyst who, knowing their own soon-to-be-published research (e.g. a strong buy recommendation) will move a stock's price, buys the stock for their own account before the report is released.

Why it is considered unethical: frontrunning based on a client's order is unethical specifically because it **directly harms the client** the broker is supposed to be acting on behalf of — the broker's own trade pushes the price against the client before the client's order is even filled, meaning the client ends up paying a higher price (or receiving a lower price, if selling) purely because of the broker's self-dealing. This is why the module's own material notes the ethics here are "more cut and dry" than ordinary insider trading (where a reasonable counterargument exists that trading on private information can improve price discovery) — frontrunning has no comparable efficiency defence, since it is a direct transfer of value from the client to the broker who is supposed to be serving that client's interests. Frontrunning is illegal in most jurisdictions when based on non-public information about pending orders or unpublished research (frontrunning based purely on public information is not illegal).

EXAM TECHNIQUE

Real, documented cases from the module's own material — Citadel Securities was fined \$700,000 for frontrunning between 2012 and 2014, and the SEC charged Sean Wygovsky with frontrunning alleging \$3.6 million in illicit profits — are strong, specific examples to cite if the question calls for one, beyond the general mechanism.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly explains the mechanism (advance position ahead of a known, price-moving transaction) AND correctly identifies the broker-client version as the clearest case AND correctly explains why it's unethical (direct harm to the client the broker should be serving, unlike ordinary insider trading's contested efficiency defence).
15-22	15-22	Correct mechanism explained, but the specific ethical reasoning (direct client harm, contrast with insider trading's efficiency defence) is only asserted.
8-14	8-14	Vague description of frontrunning as "a type of insider trading" without the specific mechanism.
0-7	0-7	No credible explanation given.

What a financial bubble is: a financial bubble occurs when an asset's price rises **far beyond its intrinsic (fundamental) value** — the value justified by the discounted stream of cash flows or utility the asset actually provides — and must, by its nature, eventually burst. As the module's own material notes (Gurkaynak, 2008), a bubble exists when "investors are willing to pay more for the stock than they know is justified by the value of the discounted dividend stream" — the key word being *know*: a bubble isn't necessarily a case of investors being fooled about the asset's true value, but of investors *knowingly* paying above that value anyway.

Is it ever rational to hold an asset you know is in a bubble? Yes — this is precisely the insight of the "rational bubbles" literature, which reframes bubbles from a symptom of investor "madness" into a coherent, profit-seeking strategy.

The Greater Fool Theory: buying a fundamentally overpriced asset is not irrational as long as there is a reasonable prospect of reselling it to someone else — the "greater fool" — at an even higher price before the bubble bursts. You are the fool for paying more than fundamental value, but if you can find a greater fool to pay you even more, the trade is still profitable.

The formal case — Blanchard and Watson (1982): in this classic model, a bubble is characterised by two possible outcomes each period: with probability q , the bubble continues to grow; with probability $(1-q)$, it collapses back to fundamental value. Crucially, the model requires the **expected return** from holding the bubble asset to properly compensate investors for the risk of collapse — the price must grow fast enough, while the bubble persists, to offset the probability-weighted loss from an eventual crash. Under this structure, holding the asset can be entirely rational and consistent with the EMH in expectation — returns and risk both rise together as the bubble inflates, exactly as ordinary asset pricing theory would require, rather than reflecting any systematic error.

The nuance — this doesn't mean holding is risk-free or costless: rational bubble models don't say holding a bubble asset is a free lunch — the higher expected return exists precisely *because* there is a real, priced-in probability of a sudden, large loss when the bubble collapses. An investor holding a rational bubble asset is knowingly accepting that risk in exchange for compensation, exactly as with any other risky asset — the difference from a "normal" asset is simply that the source of risk (a probability of sudden reversion to a much lower fundamental value) is unusually stark and, in hindsight, obvious.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly defines a bubble as a price departing from, and eventually reverting to, fundamental value AND correctly concludes holding can be rational AND explains this via both the Greater Fool Theory and the formal rational-bubbles argument (return/risk rising together, e.g. Blanchard and Watson), noting the risk is real and priced, not costless.
15-22	15-22	Correct definition and correct "yes, can be rational" conclusion, but only one of the two supporting arguments (Greater Fool / formal rational-bubbles model) given.
8-14	8-14	Correct definition, but concludes holding a known bubble asset is never rational, without engaging the rational-bubbles literature.
0-7	0-7	No credible definition or discussion given.

Framing: each measure answers a subtly different question about a portfolio's performance, and — as the module's own material stresses — the **right measure to use depends on the investor's situation**, particularly whether the portfolio being evaluated is the investor's *entire* risky holding, or just *one of several* sub-portfolios combined into a larger fund.

- **1. Sharpe Ratio:** average portfolio excess return divided by the standard deviation of returns — $(\bar{r}_p - \bar{r}_f)/\sigma_p$. It measures return per unit of **total** risk, and is also the slope of the portfolio's own Capital Allocation Line (see Q. A2(c)). It is the **appropriate measure when the portfolio represents an investor's entire risky holding**, since in that case total risk (not just systematic risk) is what actually matters to the investor.
- **2. Treynor Measure:** average portfolio excess return divided by portfolio beta — $(\bar{r}_p - \bar{r}_f)/\beta_p$. Like the Sharpe ratio, a higher value is better, but it measures excess return per unit of **systematic** risk only, on the reasoning that idiosyncratic risk should already be diversified away in a well-constructed portfolio. It is the **appropriate measure when the portfolio is just one of several sub-portfolios** combined into a larger, well-diversified fund, since in that context only the portfolio's systematic risk contributes to the overall fund's risk.

- **3. Jensen's Alpha:** unlike the two ratio measures above, alpha is not a ratio — it is the portfolio's **actual** average return minus its CAPM-**expected** return ($\alpha_p = \bar{r}_p - [\bar{r}_f + \beta_p(\bar{r}_m - \bar{r}_f)]$). A positive alpha indicates the portfolio outperformed what its systematic risk alone would predict. While originally defined relative to CAPM, it can equally be computed relative to a multifactor model such as Fama-French (see Q. A2(b)).
- **4. Information Ratio:** divides the portfolio's alpha by its **non-systematic (idiosyncratic) risk** — $\alpha_p/\sigma(\epsilon_p)$, where $\sigma(\epsilon_p)$ is sometimes called the portfolio's "tracking error." It measures abnormal return earned per unit of diversifiable risk taken on, and is the **appropriate measure when choosing an active portfolio to mix with a passive index holding**, since it directly captures how much extra, uncompensated-by-diversification risk the active manager's stock-picking is adding relative to the abnormal return it's generating.

Why the choice of measure matters in practice: the module's own worked comparison of two portfolios (P and Q) makes this concrete: if a portfolio represents an investor's *entire* investment, the Sharpe ratio (and its graphical relative, M^2) is the right comparison; if it is one of several sub-portfolios within a larger fund, the Treynor measure (or Jensen's alpha) dominates the comparison instead; and if the question is specifically which active portfolio to combine with a passive index holding, the Information Ratio is the deciding measure — the *same two portfolios* can be correctly ranked in opposite orders depending on which of these three questions is actually being asked.

Mark Allocation — Part (c) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Four distinct measures correctly named and defined with correct formulas AND each correctly distinguished by which type of risk it measures return against (total vs. systematic vs. idiosyncratic) AND the appropriate use-case for at least three of the four is correctly explained.
21-30	21-30	Four measures correctly named and defined, but the appropriate use-case distinctions are missing or only partially given.
11-20	11-20	Only two or three measures correctly defined.
0-10	0-10	Fewer than two measures correctly defined.

Question A3 Total: 100 Marks